Electromagnetic Fields Syllabus

EEE109 • Thomas Hunt

Science and Engineering • Electrical and Electronics Engineering

1. Learning Objectives & Taxonomy

ID	Learning Objective	Bloom's Level
LO1	Apply vector calculus principles to formulate fundamental electrostatic and magnetostatic equations.	Applying
LO2	Analyze electric and magnetic fields using Gauss's Law, Ampere's Law, and Biot-Savart Law for various symmetric charge and current distributions.	Analyzing
LO3	Evaluate boundary conditions, Laplace's, and Poisson's equations to solve complex electrostatic problems.	Evaluating
LO4	Examine time-varying electromagnetic fields through Faraday's Law of Induction and displacement current formulations.	Analyzing
LO5	Synthesize Maxwell's equations in both differential and integral forms to describe electromagnetic wave propagation in free space, lossy media, and good conductors.	Synthesizing
LO6	Assess the behavior of high-frequency electromagnetic signals involving wave reflection, refraction, and transmission lines.	Evaluating

Overall Course Cognitive Depth: Analyzing, Evaluating, and Synthesizing electromagnetic field theories and computational applications across Bloom's Taxonomy levels 1 through 6.

2. Weekly Instructional Plan

Wk	Subject and Activities	Outcomes
1	**Introduction to Electromagnetics & Vector Calculus Review** Face to Face Lectures, Syllabus • Discussion Forums	LO1
2	**Coulomb's Law, Electric Field Intensity, and Field Superposition** Tutorial Sessions, Face to Face Lectures • Think-Pair-Share • Group Exercises, Assignments / Homework	LO1, LO2
3	**Gauss's Law and Electric Potential** Mathematical Proofs, Face to Face Lectures • Collaborative Study Sessions • Problem Sets	LO2
4	**Conductors, Capacitance, and Energy in Electrostatic Systems** Study Questions, Face to Face Lectures • Think-Pair-Share • Concept Maps, Problem Sets	LO3
5	**Boundary Conditions and Laplace's/Poisson's Equations** Lab Projects, Face to Face Lectures • Area Specific Software • Lab Report	LO3
6	**Magnetostatic Fields: Biot-Savart Law, Ampere's Law** Real-World Cases, Face to Face Lectures • Group Discussions	LO2
7	**Magnetic Forces on Moving Charges and Charged Particle Motion** Simulations, Face to Face Lectures • Recitation Sessions • Challenge Problems, Quizzes	LO2
8	**Midterm Exam & Review of Electrostatics/Magnetostatics** Practice Exams • Tutorial Sessions • Midterm Exam	—
9	**Time-Varying Fields & Faraday's Law of Induction** Face to Face Lectures • Discussion Forums • Mathematical Proofs	LO4
10	**Displacement Current & Maxwell's Equations (Differential Form)** Study Questions, Face to Face Lectures, Lecture Notes • Quizzes	LO4, LO5
11	**Maxwell's Equations (Integral Form) & Poynting Vector** Real-World Cases, Face to Face Lectures • Group Discussions	LO5
12	**Plane Wave Propagation in Free Space and Lossless Media** Simulations, Face to Face Lectures • Lab Work • Problem Sets	LO5
13	**Wave Propagation in Lossy Dielectrics and Good Conductors (Skin Effect)** Lab Projects, Face to Face Lectures • Area Specific Software • Lab Report	LO5
14	**Reflection and Refraction of Plane Waves** Face to Face Lectures • Collaborative Study Sessions • Mathematical Proofs	LO6
15	**Introduction to Transmission Lines / Waveguides** Real-World Cases, Face to Face Lectures • Open Discussions	LO6

Wk	Subject and Activities	Outcomes
16	**Final Exam** Practice Exams, Study Questions • Office Hours, Recitation Sessions • Final Exam	—

3. Skill Acquisitions by Category

Domain	Specific Skill	Related Wk
Hard	Vector Calculus Review	1
	Electrostatic Field Calculation	2, 3
	Capacitance & Dielectric Analysis	4
	Boundary Value Problem Solving	5
	Magnetostatic Field Computation	6, 7
	Electromagnetic Induction Analysis	9
	Maxwell's Equations Application	10, 11
	Wave Propagation Simulation	12, 13
	RF and Transmission Line Analysis	14, 15
Soft	Problem Formulation	3, 7
	Collaborative Problem Solving	3, 6, 11, 14
	Technical Report Writing	5, 13
Meta	Computational Modeling	5, 13
	Critical Thinking in RF Design	12, 14, 15

4. Proposed Micro-Credential Framework

Credential Name	Dimensions	Depth	Align.	Mapping
Applied Electromagnetic Foundations	Foundational Knowledge	Proficiency	W1-4	#VectorCalculus #Electrostatics #GaussLaw #Capacitance
Computational Electromagnetics Specialist	Application	Proficiency	W5, 13	#BoundaryConditions #LaplaceEquation #SkinEffect #ComputationalModeling
RF and Wave Propagation Expert	Integration	Mastery	W10-15	#MaxwellsEquations #WavePropagation #PoyntingVector #TransmissionLines #RFDesign

5. Assessment & Mapping

Activity Name	Week	Weight	Bloom's	Outcomes
Quizzes	10	15%	Analyzing	LO4
Lab Report	5	15%	Evaluating	LO3
Midterm Exam	8	30%	Analyzing	LO1, LO2, LO3
Lab Report	13	15%	Evaluating	LO5
Final Exam	16	25%	Synthesizing	LO4, LO5, LO6